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JEE Advance Exam 2014 (Solution)

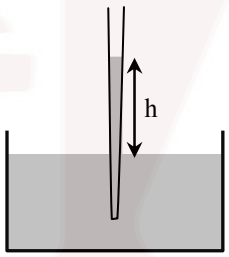
Date : 25 / 05 / 2014

Part I - PHYSICS

SECTION – 1 : (Only One Option Correct Type)

This section contains 10 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONLY ONE option is correct.

Q.1 A glass capillary tube is of the shape of a truncated cone with an apex angle α so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to a height h, where the radius of its cross section is b. If the surface tension of water is S, its density is ρ and its contact angle with glass is θ, the value of h will be (g is the acceleration due to gravity)



(A) $\frac{2S}{b\rho g}\cos(\theta - \alpha)$ (B) $\frac{2S}{b\rho g}\cos(\theta + \alpha)$

(C) $\frac{2S}{b\rho g}\cos(\theta - \alpha/2)$ (D) $\frac{2S}{b\rho g}\cos(\theta + \alpha/2)$

Ans. **[D]**

Sol. $2T\cos(\theta + \alpha/2) = \rho g b h$

$h = \frac{2S}{\rho g b}\cos(\theta + \alpha/2)$

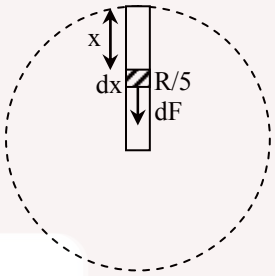
Q.2 A planet of radius R = $\frac{1}{10}$ × (radius of Earth) has the same mass density as Earth. Scientists dig a well of depth $\frac{R}{5}$ on it and lower a wire of the same length and of linear mass density 10^{-3} kgm^{-1} into it. If the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth = 6×10^6 m and acceleration due to gravity on Earth is 10 ms^{-2}).

(A) 96 N (B) 108 N (C) 120 N (D) 150 N

Ans. **[B]**

Sol. $R_P = \frac{1}{10} R_e$ $\rho_p = \rho_e$ $d_p = R_p/5$

$\lambda = 10^{-3}$ kg/m



$$g = \frac{GM}{R_e^2}$$

$$g = \frac{G}{R_e} \cdot \frac{4}{3} \pi R_e^3 \cdot \rho$$

$$g \propto R_e^2$$

$$\frac{g_p}{g_e} = \frac{R_p^2}{R_e^2}$$

$$g_p = g_e \left(\frac{1}{10}\frac{R_e}{R_e}\right)^2 = \frac{g_e}{100}$$

$$dF = dmg'$$

$$dF = \lambda dx g_p \left[1 - \frac{x}{R}\right]$$

$$dF = \lambda g_p \left[1 - \frac{x}{R_p}\right] dx$$

$$F = \lambda g_p \left[x - \frac{x^2}{2R_p}\right]$$

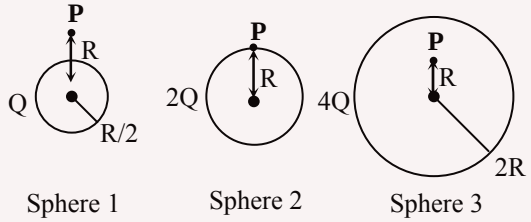
$$F = \lambda g_p \left[\frac{R_p}{5} - \frac{R_p^2}{50R_p}\right]$$

$$F = 10^{-3} \times \frac{10}{100} \left[\frac{1}{5} - \frac{1}{50}\right] \times 6 \times 10^6$$

$$= 10^{-4} \times \frac{1}{5} \times \left[1 - \frac{1}{10}\right] \times 6 \times 10^{6}$$

$$= \frac{9}{10} \times \frac{1}{5} \times 6 \times 10^{2} = 54 \times \frac{100}{50} = 108 \text{ N}$$

Q.3 Charges Q, 2Q and 4Q are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii R/2, R and 2R respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1, E_2 and E_3 respectively, then -



Sphere 1 Sphere 2 Sphere 3

(A) $E_1 > E_2 > E_3$ (B) $E_3 > E_1 > E_2$ (C) $E_2 > E_1 > E_3$ (D) $E_3 > E_2 > E_1$

Ans. **[C]**

Sol. $E_1 = k \frac{Q}{R^2}$ $E_2 = k \frac{2Q}{R^2}$ $E_3 = k \frac{4Q}{(2R)^3} . R$

Let $\frac{kQ}{R^2} = E_0$ $E_3 = K . \frac{4Q}{8R^3} R$

$E_3 = k \frac{Q}{2R^2}$

$E_1 = E_0, E_2 = 2E_0, E_3 = \frac{E_0}{2}$

$\therefore$ $E_2 > E_1 > E_3$

Q.4 If λ_{Cu} is the wavelength of K_α X-ray line of copper (atomic number 29) and λ_{Mo} is the wavelength of the K_α X-ray line of molybdenum (atomic number 42), then the ratio $\lambda_{Cu}/\lambda_{Mo}$ is close to -

(A) 1.99 (B) 2.14 (C) 0.50 (D) 0.48

Ans. **[B]**

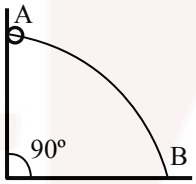
Sol. $\sqrt{\nu} = (a)(Z-1)$

$$\frac{\lambda_2}{\lambda_1} \Rightarrow \left(\frac{Z_1-1}{Z_2-1}\right)^2$$

$$\frac{\lambda_{Cu}}{\lambda_{Mo}} = \left(\frac{Z_{Mo}-1}{Z_{Cu}-1}\right)^2$$

$$= \frac{(42-1)^2}{(29-1)^2} = 2.14$$

Q.5 A wire, which passes through the hole in a small bead is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B, the force it applies on the wire is -



(A) always radially outwards

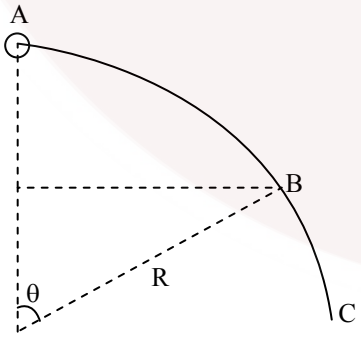
(B) always radially inwards

(C) radially outwards initially and radially inwards later

(D) radially inwards initially and radially outwards later

Ans. **[D]**

Sol.

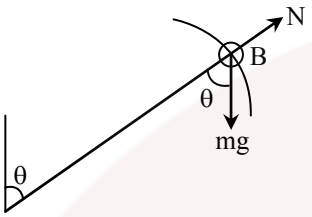


A to B

$\Delta P.E. = \frac{1}{2} mv^2$

$mgR(1 - \cos\theta) = \frac{1}{2} mv^2$ ...(1)

for circular motion at B.



$mg\cos\theta = \frac{mv^2}{R}$

$\therefore \quad mv^2 = mgR\cos\theta$ ...(2)

by (1) and (2)

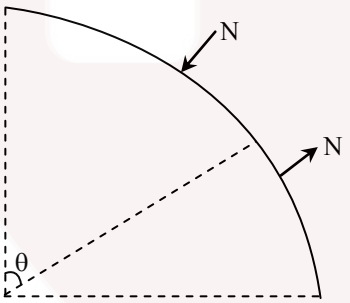
$mgR(1 - \cos\theta) = \frac{1}{2} mgR\cos\theta$

$1 = \frac{3}{2}\cos\theta$

$\cos\theta = \frac{2}{3}$

After this answer, direction of normal force act on bead is inverted.

so force on wire also inverted.



Q.6 A metal surface is illuminated by light of two different wavelengths 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2, respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240$ eV nm, the work function of the metal is nearly -

(A) 3.7 eV (B) 3.2 eV (C) 2.8 eV (D) 2.5 eV

Ans. **[A]**

Sol. Einstein photoelectric effect equation

$$KE_{max} = \frac{hc}{\lambda} - \phi$$

$$\frac{1}{2} mv_{max}^2 = \frac{hc}{\lambda} - \phi$$

$$\frac{1}{2} mu_1^2 = \frac{1240}{248} - \phi$$

$$\frac{1}{2} mu_2^2 = \frac{1240}{310} - \phi$$

$$\frac{u_1^2}{u_2^2} = \frac{\frac{1240}{248} - \phi}{\frac{1240}{310} - \phi}$$

$$\frac{4}{1} = \frac{5 - \phi}{4 - \phi}$$

$$16 - 4\phi = 5 - \phi$$

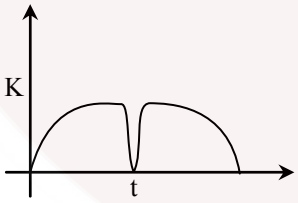
$$11 = 3\phi$$

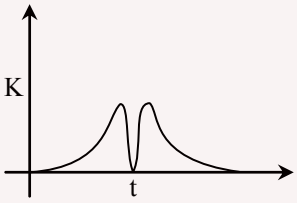
$$\phi = \frac{11}{3}$$

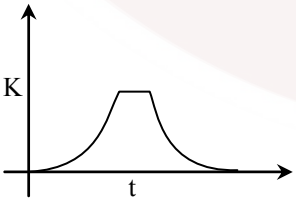
$$\Rightarrow \quad 3.66$$

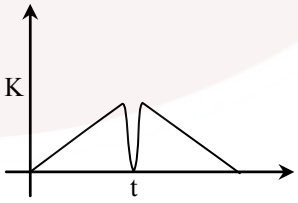
$$\Rightarrow \quad 3.7 \text{ eV}$$

Q.7 A tennis ball is dropped in a horizontal smooth surface. It bounces back to its original position after hitting the surface. The force on the ball during the collision is proportional to the length of compression of the ball. Which one of the following sketches describes the variation of its kinetic energy K with time t most appropriately ? The figures are only illustrative and not to the scale.

(A) 

(B) 

(C) 

(D) 

Ans. **[B]**

Sol. For downward journey of ball.

$v = gt$

$KE = \frac{1}{2}mv^2 = \frac{1}{2}mg^2t^2$

$KE \propto t^2$

Now ball hitting to surface

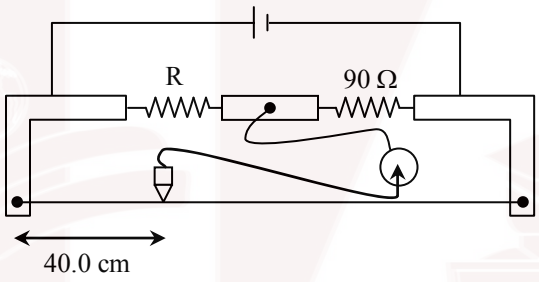
perform motion like SHM.

Starting from mean position so

$KE \propto \cos^2\omega t$

$\therefore$ graph is (B)

Q.8 During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of 90Ω, as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is -



(A) $60 \pm 0.15\Omega$ (B) $135 \pm 0.56\Omega$ (C) $60 \pm 0.25\Omega$ (D) $135 \pm 0.23\Omega$

Ans. **[C]**

Sol. $\frac{R}{90} = \frac{\ell}{100-\ell}$

$\frac{R}{90} = \frac{40}{100-40}$ $[\ell = 40 \text{ cm}]$

$R = \frac{90\times 40}{60} = 60$

$R = \frac{90\ell}{100-\ell}$

$\log R = \log 90 + \log \ell - \log (100 - \ell)$

$\frac{dR}{R} = \frac{d\ell}{\ell} - \frac{(-d\ell)}{100-\ell}$

$$\Rightarrow \quad \frac{d\ell}{\ell} + \frac{d\ell}{100-\ell}$$

$$\frac{dR}{R} \Rightarrow \frac{d\ell[100]}{(\ell)(100-\ell)}$$

$d\ell$ is least count of scale = 1 mm = 0.1 cm

$$\frac{dR}{60} = \frac{(0.1)(100)}{(40)(100-40)}$$

$$dR = \frac{60\times 0.1\times 100}{40+60}$$

$= 0.25$

$\therefore \; R = 60 \pm dR$

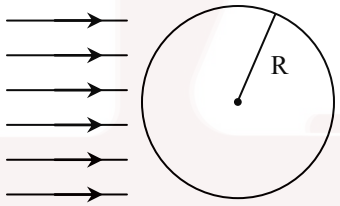
$= 60 \pm 0.25$

Q.9 Parallel rays of light of intensity I = 912 Wm^{-2} are incident on a spherical black body kept in surroundings of temperature 300 K. Take Stefan-Boltzmann constant $\sigma = 5.7 \times 10^{-8}$ Wm^{-2} K^{-4} and assume that the energy exchange with the surroundings is only through radiation. The final steady state temperature of the black body is close to -

(A) 330 K (B) 660 K (C) 990 K (D) 1550 K

Ans. **[A]**

Sol.



Rate at which energy coming on the sphere is

Flux of intensity on the sphere

Rate of energy coming = $I \times \pi R^2$ watt

Energy loss to surrounding due to radiation

$\Rightarrow \quad \sigma \times A\,(T^4 - T_s^4)$

$\Rightarrow \quad \sigma \times 4\pi R^2\,(T^4 - 300^4)$

If rate of energy coming on sphere = Rate of energy loss

then temperature of sphere become steady

$I \times \pi R^2 = \sigma \times 4\pi R^2\,(T^4 - 300^4)$

$912 = 5.7 \times 10^{-8} \times 4(T^4 - 81 \times 10^4)$

$\frac{912 \times 10^8}{5.7 \times 4} + 81 \times 10^4 = 4T^4$

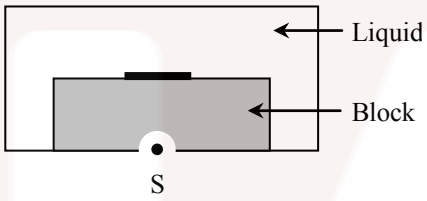
$4 \times 10^9 + 81 \times 10^4 = 4T^4$

$40000 \times 10^4 + 81 + 10^4 = 4T^4$

$\frac{400081 \times 10^4}{4} = T^4$

$T = 330\ K$

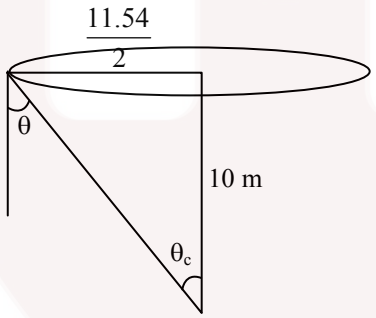
Q.10 A point source S is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is -



(A) 1.21 (B) 1.30 (C) 1.36 (D) 1.42

Ans. **[C]**

Sol.



$\mu_B = \sin\theta_C = \mu_L \sin 90^\circ$

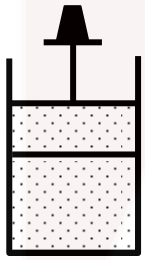
$\mu_L = 2.72 \dfrac{\frac{11.54}{2}}{\sqrt{\left(\frac{11.54}{2}\right)^2 + (10)^2}} = \dfrac{5.77}{11.54} = 1.36$

SECTION – 2 : Comprehension Type (Only One Option Correct)

This section contains 3 Paragraphs, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with two questions on each paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D).

Paragraph For Questions 11 & 12

In the figure a container is shown to have a movable (without friction) piston on top. The container and the piston are all made of perfectly insulating material allowing no heat transfer between outside and inside the container. The container is divided into two compartments by a rigid partition made of a thermally conducting material that allows slow transfer of heat. The lower compartment of the container is filled with 2 moles of an ideal monatomic gas at 700 K and the upper compartment is filled with 2 moles of an ideal diatomic gas at 400 K. The heat capacities per mole of an ideal monatomic gas are $C_V = \frac{3}{2}R$, $C_P = \frac{5}{2}R$ and those for an ideal diatomic gas are $C_V = \frac{5}{2}R$, $C_P = \frac{7}{2}R$.

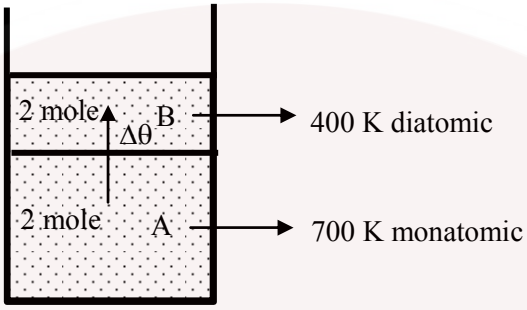


Q.11 Consider the partition to be rigidly fixed so that it does not move. When equilibrium is achieved, the final temperature of the gases will be -

(A) 550 K (B) 525 K (C) 513 K (D) 490 K

Ans. **[D]**

Sol.



$\Delta\theta_A + \Delta\theta_B = 0$

$nC_V\Delta T + nC_P\,\Delta T = 0$

$$2\times\frac{3}{2}R(T-700)+2\times\frac{7}{2}R(T-400)=0$$

$3T - 2100 + 7T - 2800 = 0$

$10T = 4900$

$T = 490$ K

Ans. (D)

Q.12 Now consider the partition to be free to move without friction so that the pressure of gases in both compartments is the same. Then total work done by the gases till the time they achieve equilibrium will be -

(A) 250 R (B) 200 R (C) 100 R (D) –100 R

Ans. **[D]**

Sol. $\Delta Q = \Delta U + \Delta W$

$0 = \Delta U + \Delta W$

$\Delta W = -\Delta U$

$\Delta W = -[nC_V\Delta T + nC_V\,\Delta T]$

$$\Delta W = -[2\times\frac{3}{2}R(T-700)+2\times\frac{5}{2}R(T-400)]$$

$= -\{3R\,(525 - 700) + 5R\,(525 - 400)\}$

$= -\{3R \times (-175) + 5R \times 125\}$

$= +525R - 625R = -100R$

Ans. (D)

$\Delta\theta_1 + \Delta\theta_2 = 0$

$nC_V\Delta T_1 + nC_P\,\Delta T_2 = 0$

$$2\times\frac{5}{2}R(T-700)+2\times\frac{7}{2}R(T-400)=0$$

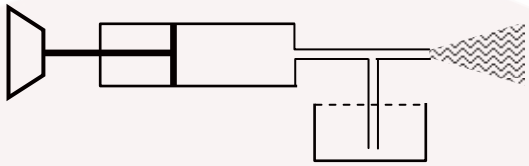
$5T - 3500 + 7T - 2800 = 0$

$12T = 6300$

$T = 525$

Paragraph For Questions 13 & 14

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere.



Q.13 If the piston is pushed at a speed of 5 mms^{-1}, the air comes out the nozzle with a speed of -

(A) 0.1 ms^{-1} (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 8 ms^{-1}

Ans. **[C]**

Sol. $A_1v_1 = A_2v_2$

$\pi d_1^2 v_1 = \pi d_2^2 v_2$

$v_2 = \left(\frac{d_1}{d_2}\right)^2 v_1$

$= \left(\frac{20}{1}\right)^2 \times 5 \text{ mms}^{-1}$

$= 400 \times 5 \times 10^{-3}$ m/s

$= 2000 \times 10^{-3}$ m/s

= 2 m/s

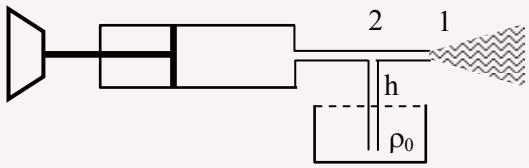
Ans. (C)

Q.14 If the density of air is ρ_a and that of the liquid ρ_ℓ, then for a given piston speed the rate (volume per unit time) at which the liquid is sprayed will be proportional to -

(A) $\sqrt{\frac{\rho_a}{\rho_\ell}}$ (B) $\sqrt{\rho_a \rho_\ell}$ (C) $\sqrt{\frac{\rho_\ell}{\rho_a}}$ (D) p_ℓ

Ans. **[A]**

Sol.



Applying Bernouli theorem between 2 & 1

$P_2 + \frac{\rho a V_a^2}{} = P_0$

$$P_2 = P_0 - \frac{\rho_a V_a^2}{2}$$

Applying Bernouli theorem between 2 & 3

$$\rho gh + \frac{\rho_\ell V_\ell^2}{2} + P_2 = P_0$$

$$\rho gh + \frac{\rho_\ell V_\ell^2}{2} + P_0 - \frac{\rho_a V_a^2}{2} P_0$$

$$V_\ell^2 = \left[\frac{\rho_a V_a^2}{2} - \rho gh\right]\frac{2}{\rho_\ell}$$

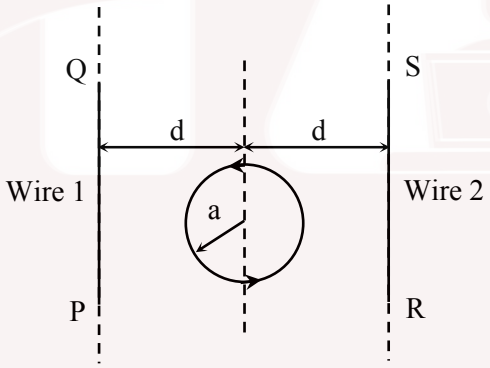
if h = 0 ; $V_\ell = \sqrt{\frac{\rho_a}{\rho_\ell}} \cdot V_a$

volume flow rate $\propto V_\ell$

Ans. (A)

Paragraph For Questions 15 & 16

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d. The loop and the wires are carrying the same current I . The current in the loop is in the counterclockwise direction if seen from above.



Q.15 When d $\approx$ a but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case

(A) current in wire 1 and wire 2 is the direction PQ and RS, respectively and h $\approx$ a

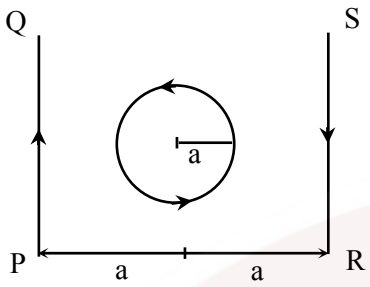
(B) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h $\approx$ a

(C) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h $\approx$ 1.2 a

(D) current in wire 1 and wire 2 is the direction PQ and RS, respectively and h $\approx$ 1.2 a

Ans. [C]

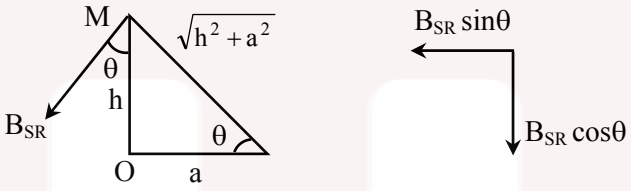
Sol.



at height h (say point M)

$$B_{loop} = \frac{\mu_0}{2}\frac{ia^2}{(h^2+a^2)^{3/2}} \odot$$

at M, Magnetic field due to loop is canceled by vertical component of magnetic field due to straight wires PQ & SR combinely.



here, $|B_{SR}| = |B_{PQ}| = \frac{\mu_0}{2\pi}\frac{i}{(h^2+a^2)^{\frac{1}{2}}} = B'$

since at M, net magnetic field is zero.

hence $|B_{loop}| = 2B'\cos\theta$

$$\frac{\mu_0}{2}\frac{ia^2}{(h^2+a^2)^{\frac{3}{2}}} = 2\times\frac{\mu_0}{2\pi}\frac{i}{(h^2+a^2)^{\frac{1}{2}}}\times\frac{a}{\sqrt{h^2+a^2}}$$

$$\boxed{h \approx 1.2a}$$

and direction of current in straight wire is along PQ and SR

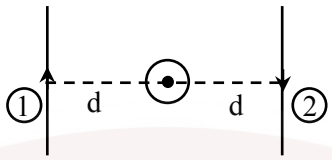
Ans. (C)

Q.16 Consider d >> a, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

(A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

Ans. **[B]**

Sol. B due to both wire at centre of loop = B_{net}



$B_{net} = \vec{B}_1 + \vec{B}_2$

$\Rightarrow \frac{\mu_0 i}{2\pi d} + \frac{\mu_0 i}{2d\pi}$

$\Rightarrow \frac{\mu_0 i}{\pi d}$

$\tau = M \times B_{Net} \times \sin 30°$

as loop is rotated by 30°

$\therefore$ angle between B_{Net} which is perpendicular to paper and M is 30°

$\tau = i \times \pi a^2 \times \frac{\mu_0 i}{\pi d} \times \sin 30°$

$\Rightarrow \frac{\mu_0 i^2 a^2}{2d}$

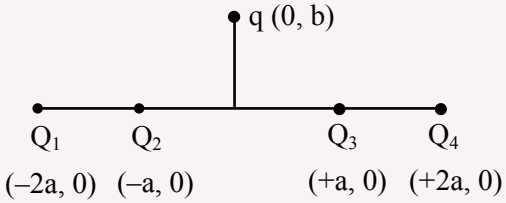
$\boxed{\tau = \frac{\mu_0 i^2 a^2}{2d}}$

Ans. (B)

SECTION – 3 : Matching List Type (Only One Option Correct)

This section contains four questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D), out of which one is correct.

Q.17 Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x-axis at x = –2a, –a, +a and +2a, respectively. A positive charge q is placed on the positive y axis at a distance b > 0. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.



	List I		List II
(P)	Q_1, Q_2, Q_3, Q_4 all positive	(1)	+ x
(Q)	Q_1, Q_2, positive; Q_3, Q_4 negative	(2)	– x
(R)	Q_1, Q_4, positive; Q_2, Q_3 negative	(3)	+ y
(S)	Q_1, Q_3, positive; Q_2, Q_4 negative	(4)	– y

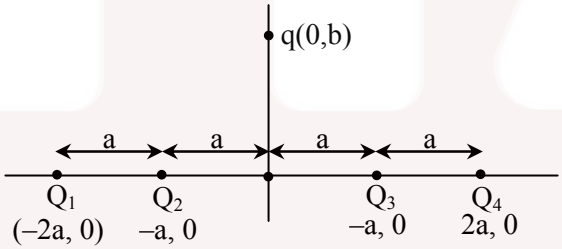
Code :

(A) P-3, Q-1, R-4, S-2 (B) P-4, Q-2, R-3, S-1

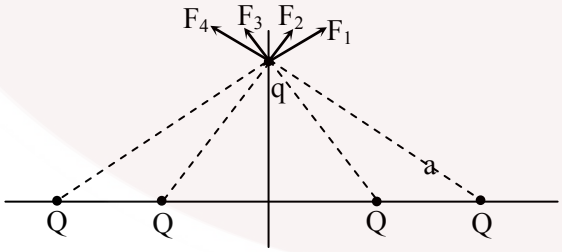
(C) P-3, Q-1, R-2, S-4 (D) P-4, Q-2, R-1, S-3

Ans. [A]

Sol.



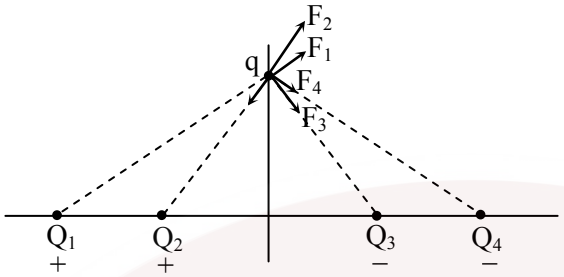
P. if all the charge are positive



F_{Net} is in +y direction

$\therefore$ P $\rightarrow$ 3

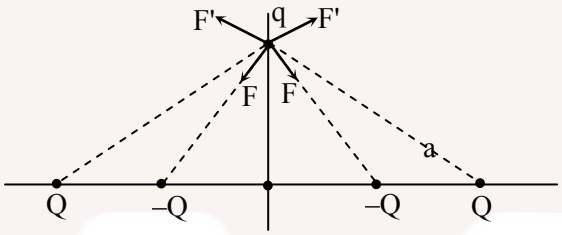
Q. if Q_1, Q_2 are positive & Q_3, Q_4 are negative



F_{Net} is in +x direction

$\therefore$ $Q \rightarrow 1$

R. If Q_1, Q_4 are positive and Q_2, Q_3 are negative



As F' < F $\therefore$ F_{Net} is in –y direction R $\rightarrow$ 4

$\therefore$ Ans. (A)

Q.18 Four combinations of two thin lenses are given in List I. The radius of curvature of all curved surfaces is r and the refractive index of all the lenses is 1.5. Match lens combinations in List I with their focal length in List II and select the correct answer using the code given below the lists.

	List I		**List II**
(P)		(1)	2r
(Q)		(2)	r/2
(R)		(3)	–r
(S)		(4)	r

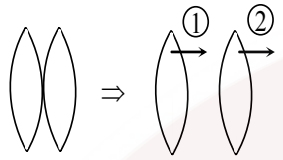
Code :

(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-4, R-3, S-1

(C) P-4, Q-1, R-2, S-3 (D) P-2, Q-1, R-3, S-4

Ans. [B]

Sol.



$$\frac{1}{f_{eq.}} = \frac{1}{f_1} + \frac{1}{f_2}$$

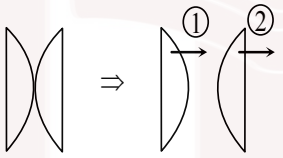
$$\frac{1}{f_1} = \left[\frac{1.5-1}{1}\right]\left[\frac{2}{r}\right] = \frac{1}{r}$$

$$\frac{1}{f_2} = \left[\frac{1.5-1}{1}\right]\left[\frac{2}{r}\right] = \frac{1}{r}$$

$$\frac{1}{f_{eq.}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{r} + \frac{1}{r} = \frac{2}{r}$$

$$f_{eq.} = \frac{r}{2}$$

$P \rightarrow 2$



$$\frac{1}{f_{eq.}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f_1} = \frac{1.5-1}{1}\left[\frac{1}{\infty} - \frac{1}{-r}\right] = \frac{0.5}{r}$$

$$\frac{1}{f_2} = \frac{1.5-1}{1}\left[\frac{1}{r} - \frac{1}{\infty}\right] = \frac{0.5}{r}$$

$$\frac{1}{f_{eq.}} = \frac{0.5}{r} + \frac{0.5}{r}$$

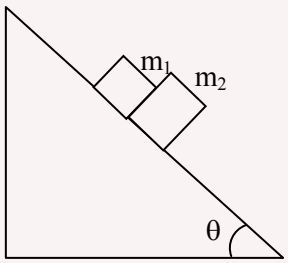
$f_{eq.} = r$

$Q \rightarrow 4$

Ans. (B)

Q.19 A block of mass m_1= 1 kg another mass m_2 = 2 kg, are placed together (see figure) on an inclined plane with angle of inclination θ. Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to μ = 0.3. In List II expressions for the friction on block m_2, are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g.

[Useful information : tan (5.5º) ≈ 0.1; tan (11.5º) ≈ 0.2; tan (16.5º) ≈ 0.3]



	List I		List II
(P)	θ = 5º	(1)	$m_2 g \sin\theta$
(Q)	θ = 10º	(2)	$(m_1 + m_2) g \sin\theta$
(R)	θ = 15º	(3)	$\mu m_2 g \cos\theta$
(S)	θ = 20º	(4)	$\mu(m_1 + m_2) g \cos\theta$

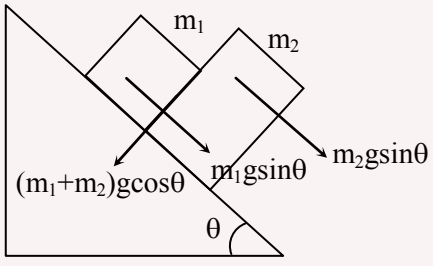
Code :

(A) P-1, Q-1, R-1, S-3 (B) P-2, Q-2, R-2, S-3

(C) P-2, Q-2, R-2, S-4 (D) P-2, Q-2, R-3, S-3

Ans. **[D]**

Sol.



if m_1 m_2 are stationary on inclined plane

then frictional force f will be static

$f = (m_1 + m_2)g\sin\theta$

this f must be less then μN

$N = m_2 g\cos\theta$

$\mu N = \mu m_2 g\cos\theta$

$f < \mu N$

$(m_1 + m_2)g\sin\theta < \mu m_2 g\cos\theta$

$(1 + 2)\sin\theta < 0.3 \times 2\cos\theta$

$\tan\theta < 0.2$

$\therefore \quad \theta < 11.5°$

for P, Q option $\theta \Rightarrow 5°$ & $10°$ which is less than $11.5°$

$\therefore$ friction $= (m_1 + m_2)g\sin\theta$

$P \rightarrow 2 \quad Q \rightarrow 2$

for option R $\theta = 15° > 11.5°$ so friction is kinetic $f = \mu N = \mu m_2 g\cos\theta$

$R \rightarrow 3$

for option S $\theta = 20°$ i.e. $\theta > 11.5°$

so $f = \mu N = \mu m_2 g\cos\theta$

$S \rightarrow 3$

Ans. (D)

Q.20 A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

	List-I		**List-II**
(P)	Lift is accelerating vertically up	(1)	d = 1.2 m
(Q)	Lift is accelerating vertically down with an acceleration less than the gravitational acceleration	(2)	d > 1.2 m
(R)	Lift is moving vertically up with constant speed	(3)	d < 1.2 m
(S)	Lift is falling freely	(4)	No water leaks out of the jar

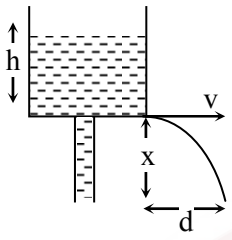
Code :

(A) P-2, Q-3, R-2, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-1, Q-1, R-1, S-4 (D) P-2, Q-3, R-1, S-1

Ans. [C]

Sol.



$x = \frac{1}{2}gt^2$

$t = \sqrt{\frac{2x}{g}}$

$v = \sqrt{2gh}$ [from Bernoulli theorem velocity efflux concept]

$d = vt$

$= \sqrt{2gh} \times \frac{\sqrt{2x}}{\sqrt{g}}$

$d = 2\sqrt{hx}$

d is independent of g or $g_{eff.}$

so whether lift is accelerating, deaccelerating or moving with constant velocity. d = 1.2 as given by examiner.

P-1, Q-1, R-1

When lift is freely falling then $g_{eff} = 0$, $v = 0$ no liquid leaks out of the jar $S \rightarrow 4$

Ans. (C)

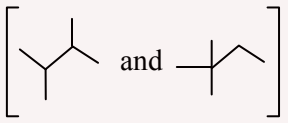
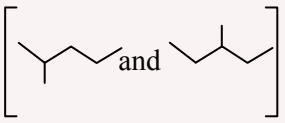
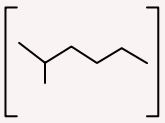
Part II - CHEMISTRY

SECTION – 1 (Only One option correct Type)

This section contains 10 **multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

Q.21 Isomers of hexane, based on their branching, can be be divided into three distinct classes as shown in the figure.

Figure

(I)  (II)  (III) 

(A) I > II > III (B) III > II > I (C) II > III > I (D) III > I > II

Ans. **[B]**

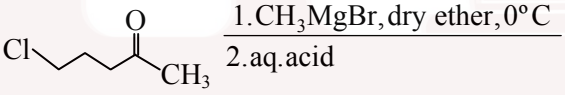
Sol. Alkane are non polar compound their boiling pt is decided by vanderwaal attraction force

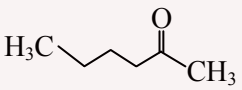
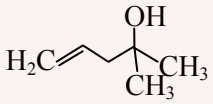
Boiling point $\propto$ Vanderwaal attraction force $\propto \frac{1}{\text{Branching}}$

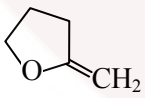
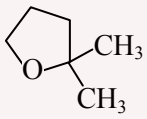
$\therefore$ Boiling point order is III > II > I

$\therefore$ Ans is (B)

Q.22 The major product in the following reaction is-

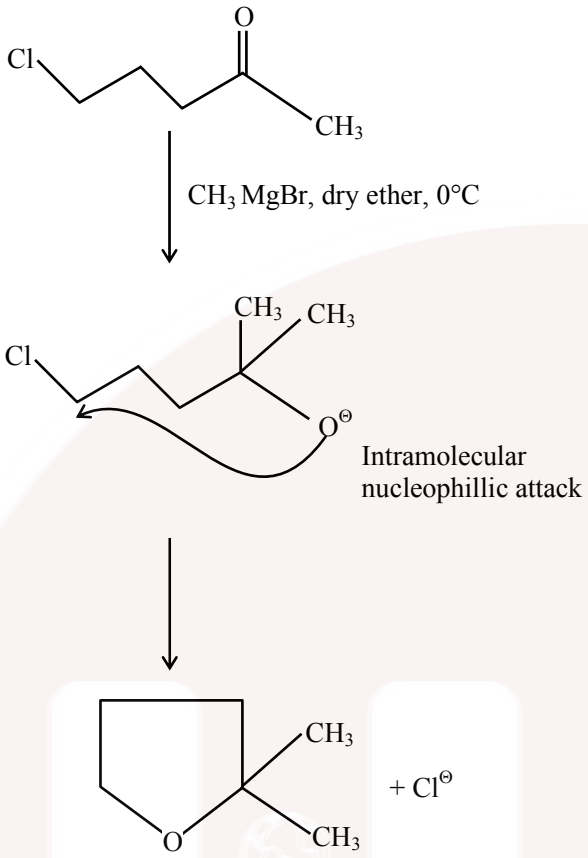


(A)  (B) 

(C)  (D) 

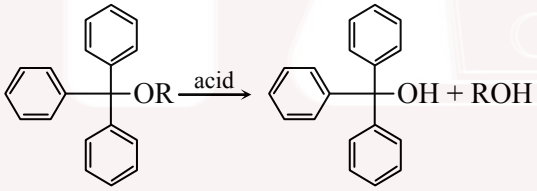
Ans. **[D]**

Sol.



Q.23 The acidic hydrolysis of ether (X) shown below is fastest when

Figure



(A) one phenyl group is replaced by a methyl group

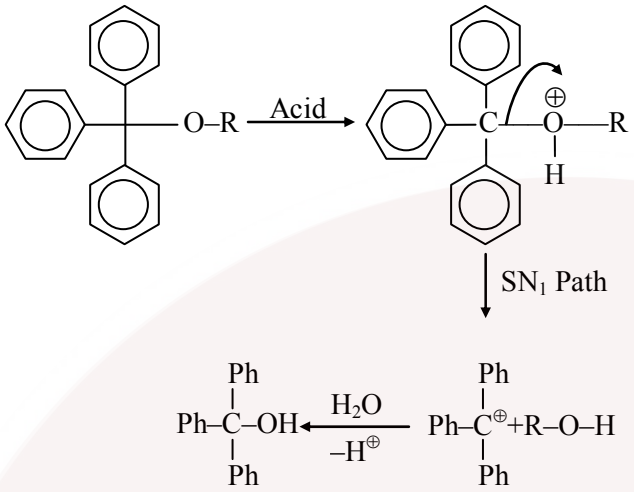
(B) one phenyl group is replaced by a *para*-methoxyphenyl group

(C) two phenyl groups are replaced by two *para*- methoxyphenyl groups

(D) no structural change is made to X.

Ans. **[C]**

Sol.



Mechanism is SN_1 so carbocation is formed so

Rate of reaction α stability of carbocation formed

so if two phenyl group are replace by two paramethoxy phenyl group then rate of reaction increases, due to +M effect of methoxy group.

Q.24 Hydrogen peroxide in its reaction with KIO_4 and NH_2OH respectively, is acting as a

(A) reducing agent, oxidising agent (B) reducing agent, reducing agent

(C) oxidising agent, oxidising agent (D) oxidising agent, reducing agent

Ans. **[A]**

Sol. $KIO_4 + H_2O_2 \rightarrow KIO_3 + O_2 + H_2O$

O.A. R.A.

$2NH_2OH + 6H_2O_2 \rightarrow 2HNO_3 + 8H_2O$

R.A. O.A.

Q.25 The product formed in the reaction of $SOCl_2$with white phosphorous is-

(A) PCl_3 (B) SO_2Cl_2 (C) SCl_2 (D) $POCl_2$

Ans. **[A]**

Sol. $P_4 + 8SOCl_2 \rightarrow 4PCl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$

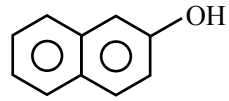
Q.26 For the identification of β-naphthol using dye test, it is necessary to use-

(A) dichloromethane solution of β-naphthol (B) acidic solution of β-naphthol

(C) neutral solution of β-napththol (D) alkaline solution of β-naphthol

Ans. **[D]**

Sol.

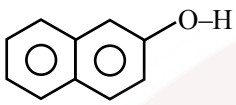


$\xrightarrow[\text{Salt}]{\text{Diazonium}}$ Orange red colour

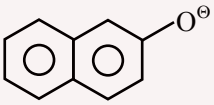
Basic medium

it should contain EDG

Basic medium is require as it will activate β-Naphthol



$\xrightarrow{OH^{\Theta}}$



+ M effect of phenoxide group

Q.27 For the process

$$H_2O(\ell) \rightarrow H_2O(g)$$

at T = 100ºC and 1 atmosphere pressure, the correct choice is

(A) $\Delta S_{system} > 0$ and $\Delta S_{urroundings} > 0$ (B) $\Delta S_{system} > 0$ and $\Delta S_{urroundings} < 0$

(C) $\Delta S_{system} < 0$ and $\Delta S_{urroundings} > 0$ (D) $\Delta S_{system} < 0$ and $\Delta S_{urroundings} < 0$

Ans. **[B]**

Sol. $H_2O_{(\ell)} \rightarrow H_2O_{(g)}$ $\Delta H > 0$ & $\Delta n_g > 0$

$\therefore$ $\Delta S_{system} > 0$ & process is phase transition

$\therefore$ $\Delta S_{surrounding} = -\Delta S_{system}$

$\therefore$ $\Delta S_{system} > 0$, $\Delta S_{surround} < 0$

i.e. $\Delta S_{system} > 0$ & $\Delta S_{surrounding} < 0$

Q.28 For the elementary reaction **M** $\rightarrow$ **N**, the rate of disappearance of **M** increases by a factor of 8 upon doubling the concentration of **M**. The order of the reaction with respect to **M** is-

(A) 4 (B) 3 (C) 2 (D) 1

Ans. **[B]**

Sol. $\frac{-d[M]}{dt} = r = k\,[M]^x$ ……(i)

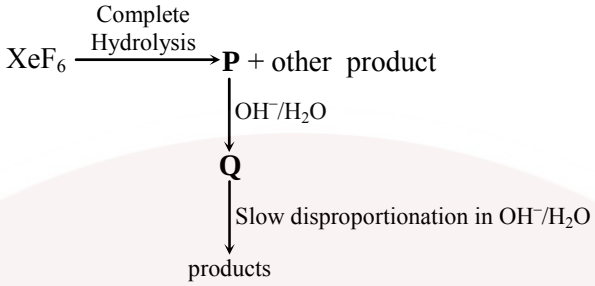
$8r = k\,(2M)^2$ …..(ii)

(i) ÷ (ii)

$\frac{1}{2^3} = \frac{1}{2^x}$

x = 3 order.

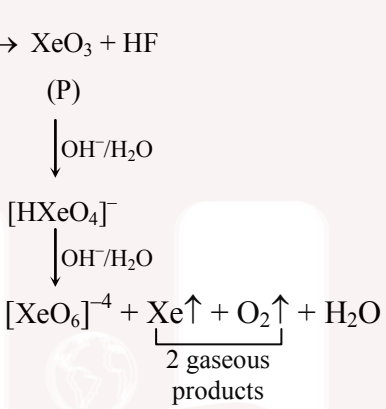
Q.29 Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is-



(A) 0 (B) 1 (C) 2 (D) 3

Ans. **[C]**

Sol.



Q.30 Assuming 2s-2p mixing is **NOT** operative, the paramagnetic species among the following is-

(A) Be_2 (B) B_2 (C) C_2 (D) N_2

Ans. **[C]**

Sol. Assuming 2s-2p mixing is not operative, then energy level diagram becomes

$\sigma^*_{2p_z}$

$\pi^*_{2p_x} = \pi^*_{2p_y}$

$\pi_{2p_x} = \pi_{2py}$

$\sigma^*_{2p_z}$

σ^*_{2s}

σ_{2s}

σ^*_{1s}

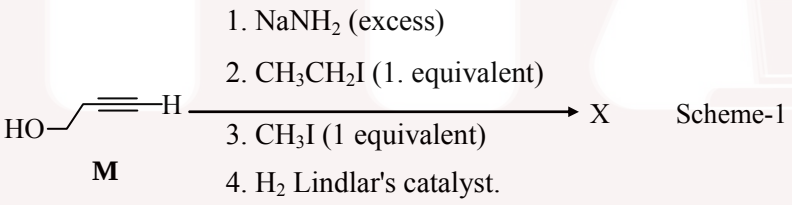
σ_{1s}

SECTION – 2 : Comprehension Type (Only One Option Corect)

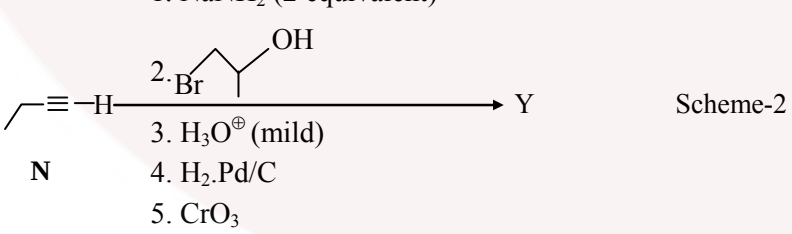
This section contains **3 paragraphs**, each describing theory, experiments, data etc. 6 questions relate to the three paragraphs with two questions on each paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D)

Paragraph For Questions 31 and 32

Schemes 1 and 2 describe sequential transformation of alkynes M and N. Consider only the major products formed in each step for both the sechemes.

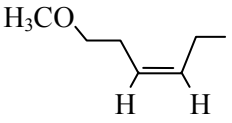


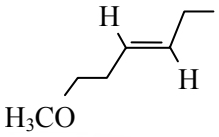
M Scheme-1

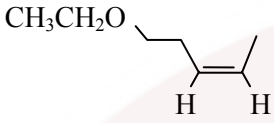


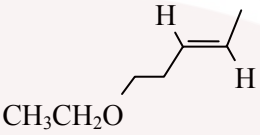
N Scheme-2

Q.31 The product X is -

(A) 

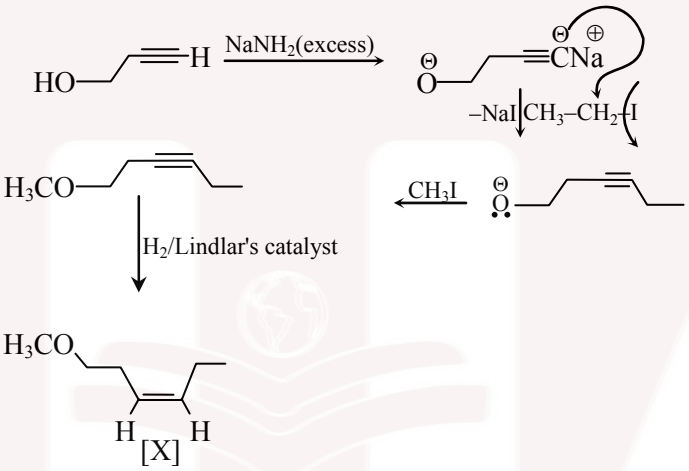
(B) 

(C) 

(D) 

Ans. **[A]**

Sol.



Q.32 The correct statement with respect to product Y is -

(A) It gives a positive Tollens test and is a functional isomer of X

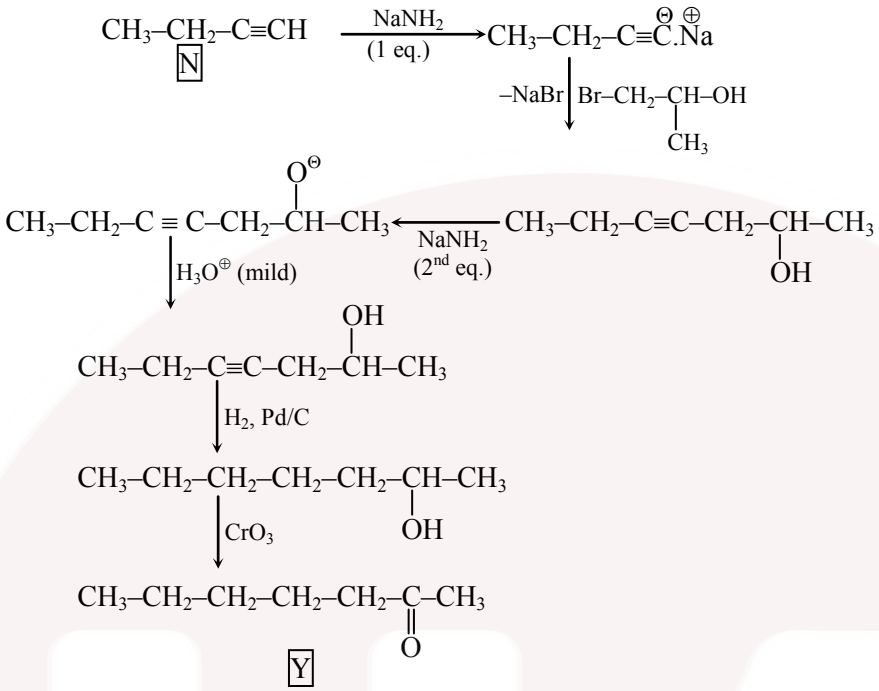
(B) It gives a positive Tollens test and is a geometrical isomer of X

(C) It gives a positive iodoform test and is a functional isomer of X

(D) It gives a positive iodoform test and is a geomtrical isomer of X

Ans. **[C]**

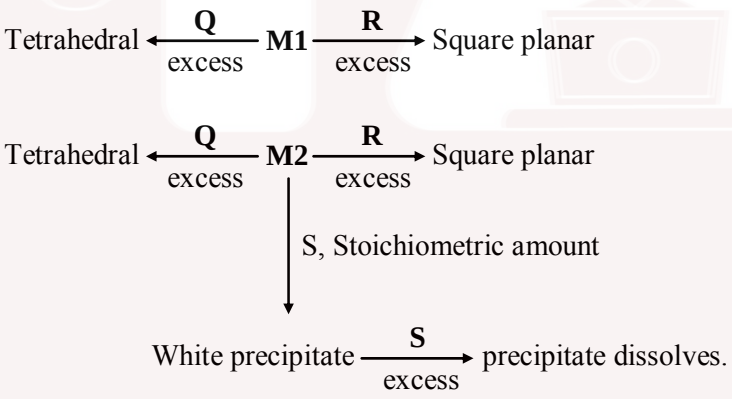
Sol.



Paragraph For Questions 33 and 34

An aqueous solution of metal ion M1 reacts seaprately with reagents Q and R in excess to give tetrahedral and square planar complexes, An aqueous solution of another metal ion M2 always forms tetrahedral complexes with these reagents. Aqueous solution of M2 on reaction with reagent S gives white precipite which dissolves in excess of S. The reactions are summarised in The scheme given below.

SCHEME :



Q.33 M1, Q and R respectively are -

(A) Zn^{2+}, KCN and HCl (B) Ni^{2+}, HCl and KCN

(C) Cd^{2+}, KCN and HCl (D) Co^{2+}, HCl and KCN

Ans. **[B]**

Sol. $H_2[NiCl_4] \xleftarrow[\text{excess}]{HCl(Q)} Ni^{+2} \xrightarrow[\text{excess}]{KCN(R)} K_2[Ni(CN)_4]$

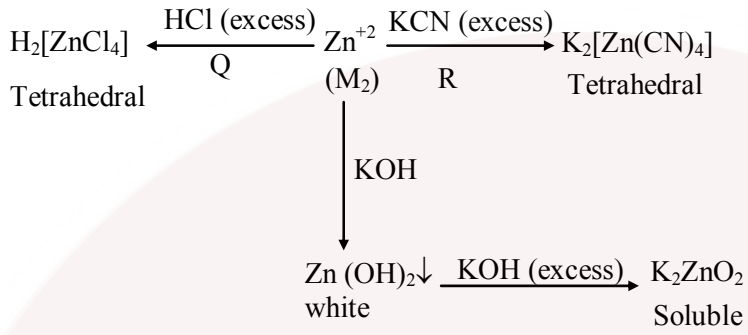
Tetrahedral (M_1) square planar

Q.34 Reagent **S** is-

(A) $K_4[Fe(CN)_6]$ (B) Na_2HPO_2 (C) K_2CrO_4 (D) KOH

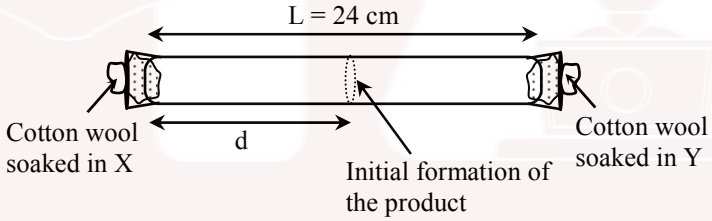
Ans. **[D]**

Sol.



Paragraph For Questions 35 and 36

X and **Y** are two volatile liquids with molar weights of 10g mol^{-1} and 40g mol^{-1} respectively. Two cotton plugs, one soaked in **X** and the other soaked in **Y**, are simultaneously placed at the ends of a tube of length **L** = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300 K. Vapours of **X** and **Y** react to form a product which is first observed at a distance **d** cm from the plug soaked in **X**. Take **X** and **Y** to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.



Q.35 The value of **d** in cm (shown in the figure), as estimated from Graham's law, is

(A) 8 (B) 12 (C) 16 (D) 20

Ans. **[C]**

Sol. $\frac{R_x}{R_y} = \sqrt{\frac{M_y}{M_x}}$ {for same T & P}

$$\frac{A \times d \times t}{t \times A \times (24-d)} = \sqrt{\frac{M_y}{M_x}}$$

$$\frac{d}{24-d}=\sqrt{\frac{40}{10}}$$

$$\frac{d}{24-d}=2$$

$$d = 48 - 2d$$

$$3d = 48$$

$$d=\frac{48}{3}=16\,cm$$

Q.36 The experimental value of **d** is found to be smaller than the estimate obtained using Graham's law. This is due to

(A) larger mean free path for **X** as compared to that of **Y**.

(B) larger mean free path for **Y** as compared to that of **X**.

(C) increased collision frequency of **Y** with the inert gas as compared to that of **X** with the inert gas.

(D) increased collision frequency of **X** with the inert gas as compared to that of **Y** with the inert gas.

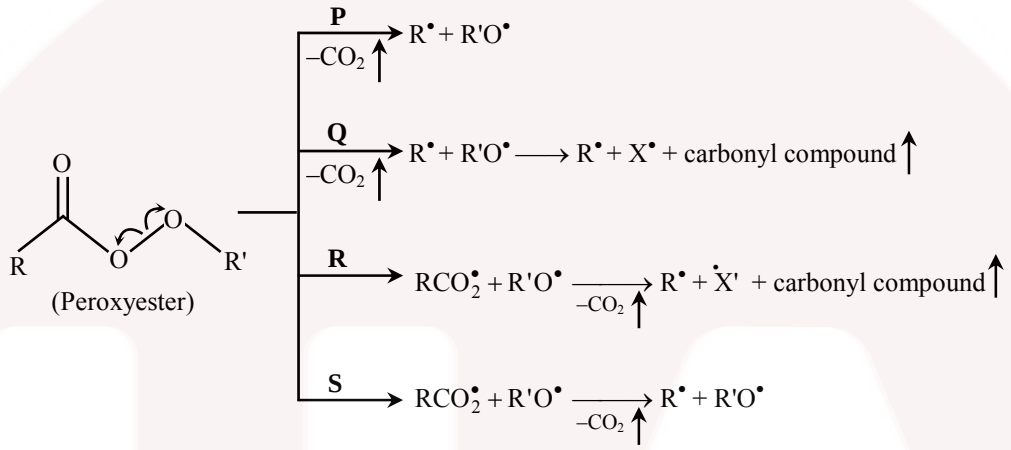
Ans. **[D]**

Sol. Greater is the mass smaller is the collision frequency & hence larger is the distance travelled.

SECTION – 3 : Matching List Type (Only One Option Correct)

This section contains four questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D), out of which one is correct.

Q.37 Different possible **thermal** decomposition pathways for peroxyesters are shown below. Match each pathway from **List-I** with an appropriate structure from **List-II** and select the correct answer using the code given below the lists



List – I

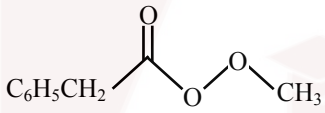
P. Pathway **P**

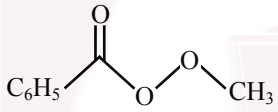
Q. Pathway **Q**

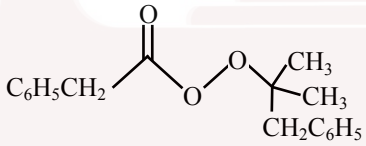
R. Pathway **R**

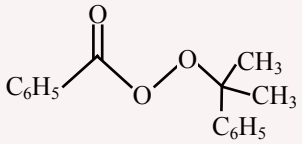
S. Pathway **S**

List – II

1. 

2. 

3. 

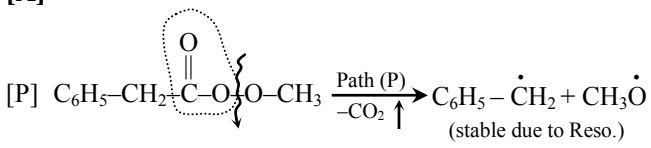
4. 

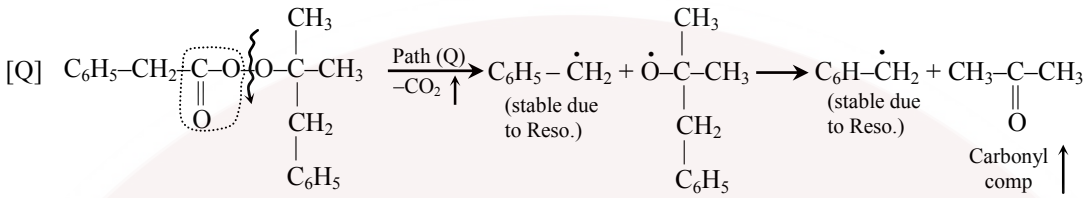
Codes :

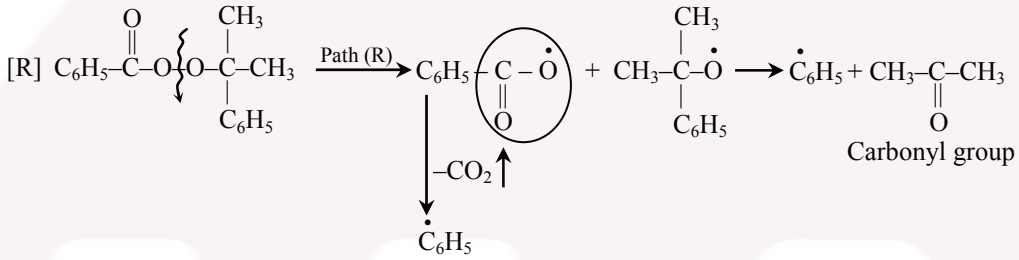
	P	Q	R	S
(A)	1	3	4	2
(B)	2	4	3	1
(C)	4	1	2	3
(D)	3	2	1	4

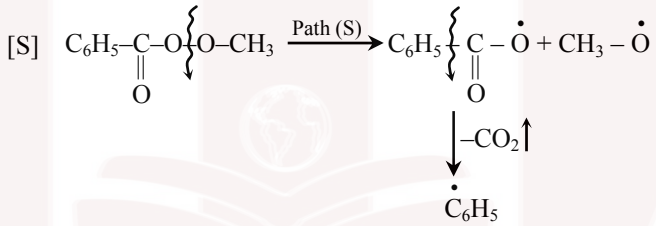
Ans. **[A]**

Sol.

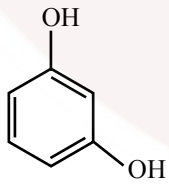


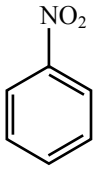




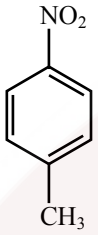


Q.38 Match the four starting materials (**P, Q, R, S**) given in **List-I** with the corresponding reaction schemes (I, II, III, IV) provided in **List-II** and select the correct answer using the code given below the lists.

List – I	**List – II**
P. H—≡—H	**1. Scheme I** (i) $KMnO_4$. $HO^{\ominus}$. heat (ii) $H^{\oplus}$, H_2O (iii) $SOCl_2$ (iv) NH_3 ? → $C_7H_6N_2O_3$
Q. 	**2. Scheme II** (i) Sn/HCl (ii) CH_3COCl (iii) conc.H_2SO_4 (iv) HNO_3 (v) dil.H_2SO_4, heat (vi) $HO^{\ominus}$? → $C_6H_6N_2O_2$

R. 

3. **Scheme III**
(i) red hot iron, 873 K (ii) fuming HNO_3, H_2SO_4, heat
(iii) $H_2S.NH_3$ (iv) $NaNO_2$, H_2SO_4 (v) hydrolysis
? $\longrightarrow$ $C_6H_5NO_3$

S. 

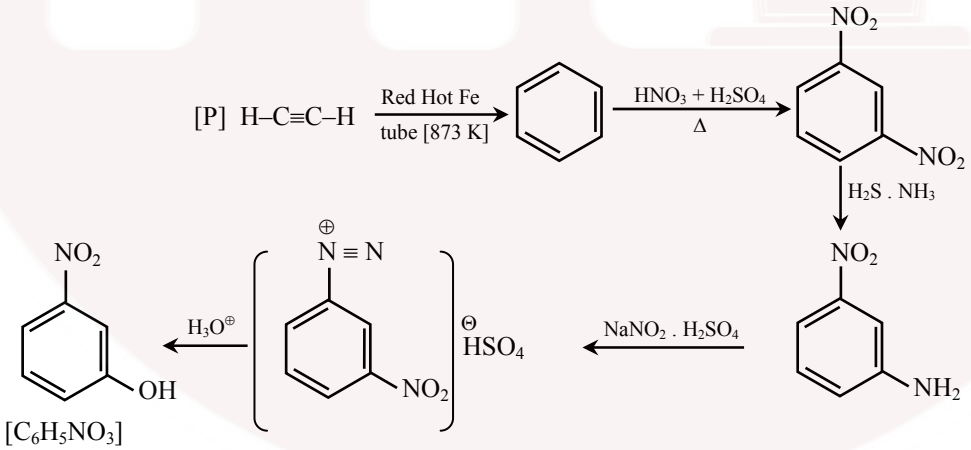
4. **Scheme IV**
(i) conc. H_2SO_4, 60°C
(ii) conc. HNO_3, conc. H_2SO_4 (iii) dil. H_2SO_4, heat
? $\longrightarrow$ $C_6H_5NO_4$

Codes :

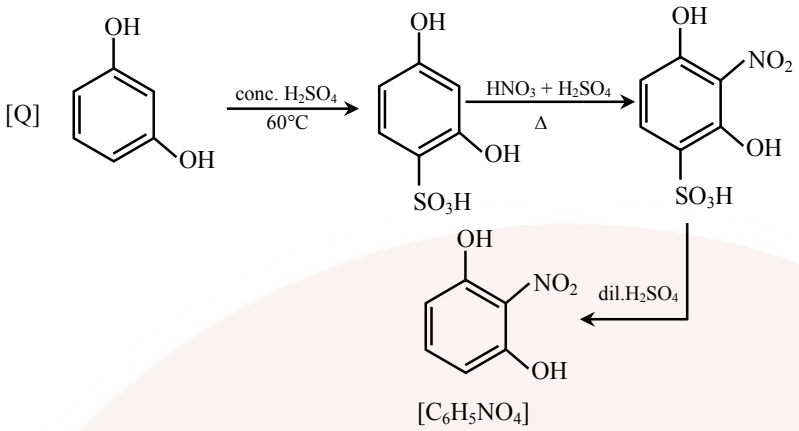
	P	Q	R	S
(A)	1	4	2	3
(B)	3	1	4	2
(C)	3	4	2	1
(D)	4	1	3	2

Ans. **[C]**

Sol.

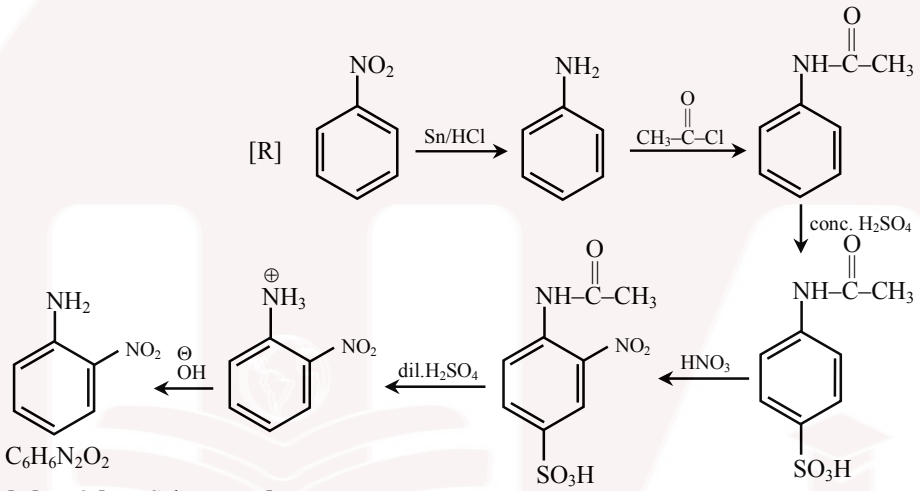


[P] $\rightarrow$ 3 [P $\rightarrow$ Scheme - III]



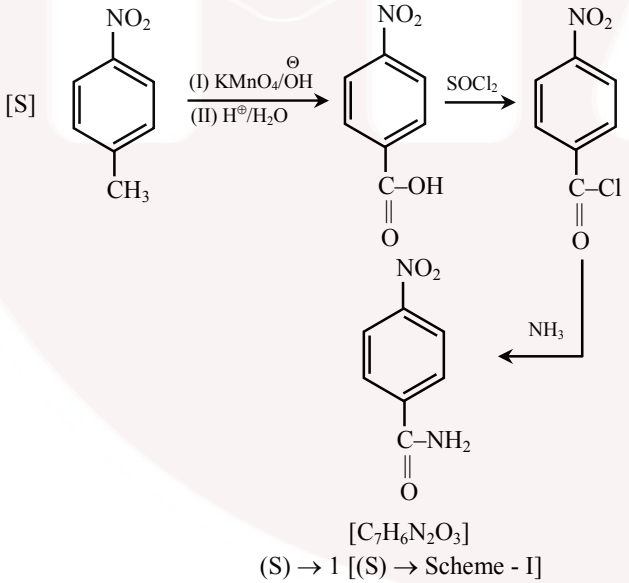
$[C_6H_5NO_4]$

(Q) → 4 [Q → Scheme - N]



$C_6H_6N_2O_2$

[R] → 2 [R – Scheme - II]



$[C_7H_6N_2O_3]$

(S) → 1 [(S) → Scheme - I]

Smart Trick: Student can think about no. of carbon atom in reactant side as well as product side and able to give correct answer in this question.

Q.39 Match each coordination compound in **List-I** with an appropriate pair of characteristics from **List-II** and select the correct answer using the code given below the lists.

{en = $H_2NCH_2CH_2NH_2$: atomic numbers ; Ti = 22; Cr = 24; Co = 27; Pt = 78}

List – I		**List – II**	
(P)	$[Cr(NH_3)_4Cl_2]Cl$	**(1)**	Paramagnetic and exhibits ionization isomerism
(Q)	$[Ti(H_2O)_5Cl](NO_3)_2$	**(2)**	Diamagnetic and exhibits cis -trans isomerism
(R)	$[Pt(en)(NH_3)Cl]NO_3$	**(3)**	Paramagnetic and exhibits cis-trans isomerism
(S)	$[Co(NH_3)_4(NO_3)_2]NO_3$	**(4)**	Diamagnetic and exhibits ionization isomerism

Codes :

	P	**Q**	**R**	**S**
(A)	4	2	3	1
(B)	3	1	4	2
(C)	2	1	3	4
(D)	1	3	4	2

Ans. **[B]**

Sol. [P] $[Cr(NH_3)_4Cl_2]Cl$

Cr^{+3} = [Ar] $4s^\circ$ $3d^3$

It has 3 unpaired e^- . Hence,

This complex is paramagnetic Ma_4b_2 complex can show cis-trans isomerism

[Q] $[Ti(H_2O)_5Cl](NO_3)_2$

Ti^{+3} = [Ar] $4s^\circ$ $3d^1$

It has 1 unpaired e^-. Hence,

Complex is paramagnetic. It can show ionisation isomerism

[R] $[Pt(en)(NH_3)Cl]NO_3$

It has zero unpaired e^- . Hence

Complex is diamagnetic and show ionisation isomerism

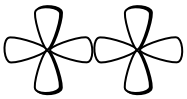
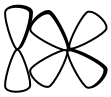
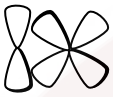
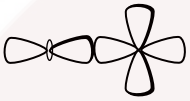
[S] $[Co(NH_3)_4(NO_3)_2]NO_3$

Co^{+3} = [Ar] $4s^\circ$ $3d^6$

Due to presence of strong field ligands, zero unpaired e^- is present. Hence, complex is diamagnetic.

Ma_4b_2 complex can show cis-trans isomerism.

Q.40 Match the orbital overlap figures shown in **List-I** with the description given in **List-II** and select the correct answer using the code given below the lists.

List – I		**List – II**	
(P)		**(1)**	p – d π antibonding
(Q)		**(2)**	d – d σ bonding
(R)		**(3)**	p – d π bonding
(S)		**(4)**	d – d σ antibonding

Codes :

	P	**Q**	**R**	**S**
(A)	2	1	3	4
(B)	4	3	1	2
(C)	2	3	1	4
(D)	4	1	3	2

Ans. **[C]**

Sol. [P] → d – d σ bonding

[Q] → p – d π bonding

[R] → p – d π anti bonding

[S] → d – d σ anti bonding

Part III - MATHEMATICS

SECTION – 1 (Only One option correct Type)

This section contains 10 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Q.41 Coefficient of x^{11} in the expansion of $(1 + x^2)^4 (1 + x^3)^7 (1 + x^4)^{12}$ is

(A) 1051 (B) 1106 (C) 1113 (D) 1120

Ans. **[C]**

Sol. Coefficient of x^{11} in $(1 + x^2)^4 (1 + x^3)^7 (1 + x^4)^{12}$

Now consider the following cases

	$(1 + x^2)^4$	$(1 + x^3)^7$	$(1 + x^4)^{12}$
Coefficient of	x^0	x^3	x^8
Coefficient of	x^2	x^9	x^0
Coefficient of	x^4	x^3	x^4
Coefficient of	x^8	x^3	x^0

$= {}^4C_0 \times {}^7C_1 \times {}^{12}C_2 + {}^4C_1 \times {}^7C_3 \times {}^{12}C_0 + {}^4C_2 \times {}^7C_1 \times {}^{12}C_1 + {}^4C_4 \times {}^7C_1 \times {}^{12}C_0$

$= 462 + 140 + 504 + 7 = 1113$

Q.42 Let $f : [0, 2] \to R$ be a function which is continuous on [0, 2] and is differentiable on (0, 2) with $f(0) = 1$. Let

$F(x) = \int_0^{x^2} f(\sqrt{t})\, dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then F(2) equals

(A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

Ans. **[B]**

Sol. $F'(x) = f'(x) \Rightarrow 2x.f(x) = f'(x) \Rightarrow \frac{f'(x)}{f(x)} = 2x$

$\Rightarrow \int \frac{f'(x)}{f(x)}\, dx = \int 2x\, dx \Rightarrow \ell n\, f(x) = x^2 + c$

$\Rightarrow f(x) = e^{x^2+c} \quad \Rightarrow f(x) = k\, e^{x^2}$

as $f(0) = 1 \Rightarrow k = 1$

$\therefore f(x) = e^{x^2}$

$\Rightarrow F(x) = e^{x^2} - e^0 = e^{x^2} - 1 \Rightarrow F(2) = e^4 - 1$

Q.43 The function y = f(x) is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in (–1, 1) satisfying f(0) = 0. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\,dx$ is

(A) $\frac{\pi}{3}-\frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6}-\frac{\sqrt{3}}{2}$

Ans. **[B]**

Sol. $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$

I.F. $= e^{\int \frac{xdx}{x^2-1}} = e^{\frac{1}{2}\ln|x^2-1|}$

$= \sqrt{1-x^2}$ (as $x^2 < 1$)

So solution is

$y\sqrt{1-x^2} = \int \frac{x^4+2x}{\sqrt{1-x^2}} \times \sqrt{1-x^2}\,dx$

$y\sqrt{1-x^2} = \frac{x^5}{5} + x^2 + c$

as it passes through (0, 0) so c = 0

$\therefore\ f(x) = \frac{\frac{x^5}{5}+x^2}{\sqrt{1-x^2}}$

$\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx = \frac{1}{5}\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^5}{\sqrt{1-x^2}}\,dx + \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx$

$= 0 + 2\int_0^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}\,dx = 2\int_0^{\frac{\sqrt{3}}{2}} \left(\frac{1}{\sqrt{1-x^2}} - \sqrt{1-x^2}\right) dx$

$= 2\left(\sin^{-1}x\right)_0^{\frac{\sqrt{3}}{2}} - 2\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_0^{\frac{\sqrt{3}}{2}} = 2\times\frac{\pi}{3} - 2\left[\frac{\sqrt{3}}{4}\times\frac{1}{2} + \frac{1}{2}\times\frac{\pi}{3}\right]$

$= \frac{2\pi}{3} - \frac{\sqrt{3}}{4} - \frac{\pi}{3} = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$

Q.44 The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\operatorname{cosec} x)^{17}\, dx$ is equal to

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16}\, du$ (B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17}\, du$

(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17}\, du$ (D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16}\, du$

Ans. **[A]**

Sol. $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\operatorname{cosec} x)^{17}\, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{2^{17}(\operatorname{cosec} x)^{16} \operatorname{cosec} x(\operatorname{cosec} x + \cot x)}{(\operatorname{cosec} x + \cot x)} dx$

Let $\operatorname{cosec} x + \cot x = t$

$(-\operatorname{cosec} x \cdot \cot x - \operatorname{cosec}^2 x)dx = dt$

$$\operatorname{cosec} x - \cot x = \frac{1}{t}$$

$$= -\int_{\sqrt{2}+1}^{1} 2^{17} \left(\frac{t + \frac{1}{t}}{2} \right)^{16} \frac{dt}{t}$$

Let $t = e^u \Rightarrow dt = e^u\, du$

$$= -\int_{\ell n(\sqrt{2}+1)}^{0} 2(e^u + e^{-u})^{16} \frac{e^u du}{e^u}$$

$$= 2\int_0^{\ell n(\sqrt{2}+1)} (e^u + e^{-u})^{16}\, du$$

Q.45 The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has

(A) only purely imaginary roots (B) all real roots

(C) two real and two purely imaginary roots (D) neither real nor purely imaginary roots

Ans. **[D]**

Sol. Let roots of $p(x) = 0$ are $ki, -ki$, where $k \in R$

$\alpha + \beta = 0, \alpha\beta = k^2$

$\therefore\ p(x) = x^2 + k^2$

$p(p(x)) = (x^2 + k^2)^2 + k^2 = 0$

$x^4 + 2x^2 k^2 + k^4 + k^2 = 0$

$$x^2 = \frac{-2k^2 \pm \sqrt{4k^4 - 4k^4 - 4k^2}}{2}$$

$$x^2 = -k^2 \pm ki$$

$$x = \pm \sqrt{-k^2 \pm ki}$$

So there will be 4 imaginary roots but none of them is purely real or purely imaginary.

Q.46 For $x \in (0, \pi)$, the equation $\sin x + 2 \sin 2x - \sin 3x = 3$ has

(A) infinitely many solutions (B) three solutions

(C) one solution (D) no solution

Ans. **[D]**

Sol. $\sin x - \sin 3x + 2 \sin 2x = 3$

$\Rightarrow -2 \cos 2x \sin x + 4 \sin x \cos x = 3$

$\Rightarrow 2 \sin x (2 \cos x - \cos 2x) = 3$

$\Rightarrow 2 \sin x (2 \cos x - 2 \cos^2 x + 1) = 3$

$$\Rightarrow 2 \sin x \left(\frac{3}{2} - 2\left(\cos x - \frac{1}{2} \right)^2 \right) = 3$$

$$\Rightarrow 3 \sin x - 3 = 4\left(\cos x - \frac{1}{2} \right)^2 \sin x$$

as $x \in (0, \pi)$

LHS ≤ 0 and RHS ≥ 0

But if L.H.S = R.H.S. = 0 then no common value of x is satisfied

So no solution of the equation.

Q.47 In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

Ans. **[B]**

Sol. $a + b = x$, $ab = y$, $2s = x + c$

and $x^2 - c^2 = y$

Now $\frac{r}{R} = \frac{\Delta/s}{abc/4\Delta} = \frac{4\Delta^2}{sabc} = \frac{4s(s-a)(s-b)(s-c)}{s\,abc}$

$= \frac{4(s^2 - s(a+b) + ab)(s-c)}{abc}$

$= \frac{((2s)^2 - 2(a+b)(2s) + 4ab)(s-c)}{abc}$

$= \frac{((x+c)^2 - 2x(x+c) + 4y)\left(\frac{x+c}{2} - c\right)}{y\,c}$

$= \frac{(-x^2 + c^2 + 4y)(x-c)}{2cy}$

$= \frac{3y(x-c)}{2cy} = \frac{3(x-c)}{2c} = \frac{3y}{2c(x+c)}$

Q.48 Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. **[A]**

Sol. Total number of ways to arrange three boys and two girls are 5!.

According to given condition following cases may arise

1	2	3	4	5
G	G	B	B	B
G	B	B	G	B
G	B	B	B	G
B	G	B	B	G
B	G	G	B	B

So, Number of favourable ways = $5 \times 3! \times 2!$

$= 60$

$\therefore$ Required probability = $\frac{60}{120} = \frac{1}{2}$

Q.49 The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is

(A) 3 (B) 6 (C) 9 (D) 15

Ans. **[D]**

Sol. Let equation of tangent to parabola is $y = mx + \frac{2}{m}$

It also touches circle $x^2 + y^2 = 2$

$$\left|\frac{2}{m\sqrt{1+m^2}}\right| = \sqrt{2}$$

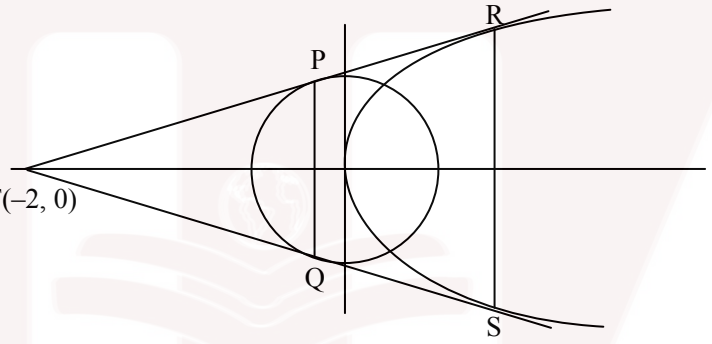
$\Rightarrow m^4 + m^2 = 2$

$\Rightarrow m^4 + m^2 - 2 = 0$

$\Rightarrow (m^2 - 1)(m^2 + 2) = 0 \Rightarrow m = \pm 1$

So tangents are $y = x + 2$, $y = -x - 2$

they intersects at $(-2, 0)$



Equation of chord PQ is $(-2)(x) - 2 = 0 \Rightarrow x = -1$

Equation of chord RS is $-4(x - 2) = 0 \Rightarrow x = 2$

So, coordinates of P,Q,R,S are P(–1, 1), Q(–1, –1), R(2, 4), S(2, – 4)

Hence area = $\frac{(8+2)\times 3}{2}$ = 15 square units

Q.50 Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is

(A) 264 (B) 265 (C) 53 (D) 67

Ans. **[C]**

Sol. Let we have letter $L_1, L_2, L_3, L_4, L_5, L_6$

and Envelopes $E_1, E_2, E_3, E_4, E_5, E_6$

If $L_1 \rightarrow E_2 = x$

$L_1 \to E_3 = x$

$L_1 \to E_4 = x$

$L_1 \to E_5 = x$

$L_1 \to E_6 = x$

Total $= 5x = D_6$

$$D_6 = 6!\left[\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right] = 265$$

$\therefore\ x = \dfrac{265}{5} = 53$

SECTION – 2 : Comprehension Type (Only One Option Correct)

This section contains 3 **paragraphs**, each describing theory, experiments, data etc. Six questions relate to the three paragraphs with two questions on each paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D).

Paragraph For Questions 51 and 52

Let a, r, s, t be nonzero real numbers. Let P(at^2, 2at), Q, R(ar^2, 2ar) and S(as^2, 2as) be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point (2a, 0).

Q.51 The value of r is

(A) $-\dfrac{1}{t}$ (B) $\dfrac{t^2+1}{t}$ (C) $\dfrac{1}{t}$ (D) $\dfrac{t^2-1}{t}$

Ans. **[D]**

Sol. If PQ is focal chord then coordinates of Q will be $\left(\dfrac{a}{t^2}, -\dfrac{2a}{t}\right)$.

Now slope of QR = slope of PK

$$\frac{2ar+\frac{2a}{t}}{ar^2-\frac{a}{t^2}} = \frac{2at}{at^2-2a}$$

$$\Rightarrow \frac{r+\frac{1}{t}}{r^2-\frac{1}{t^2}} = \frac{t}{t^2-2} \quad \Rightarrow \frac{1}{r-\frac{1}{t}} = \frac{t}{t^2-2}$$

$$\Rightarrow r-\frac{1}{t} = \frac{t^2-2}{t} = t-\frac{2}{t} \Rightarrow r = t-\frac{1}{t} = \frac{t^2-1}{t}$$

Q.52 If st = 1, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is

(A) $\frac{(t^2+1)^2}{2t^3}$ (B) $\frac{a(t^2+1)^2}{2t^3}$ (C) $\frac{a(t^2+1)^2}{t^3}$ (D) $\frac{a(t^2+2)^2}{t^3}$

Ans. **[B]**

Sol. Equation of tangent at P is $ty = x + at^2$ …(i)

Equation of normal at S is

$y = -sx + 2as + as^3$

as $s = \frac{1}{t}$

$y = -\frac{x}{t} + \frac{2a}{t} + \frac{a}{t^3}$ …(ii)

from (i) $x = ty - at^2$

Now $y = -\frac{(ty - at^2)}{t} + \frac{2a}{t} + \frac{a}{t^3}$

$2y = at + \frac{2a}{t} + \frac{a}{t^3}$

$y = \frac{a}{2}\left(\frac{t^4 + 2t^2 + 1}{t^3}\right) = \frac{a(t^2+1)^2}{2t^3}$

Paragraph For Questions 53 and 54

Given that for each $a \in (0, 1)$,

$$\lim_{h\to 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1}dt$$

exists. Let this limit be g(a). In addition, it is given that the function g(a) is differentiable on (0,1).

Q.53 The value of $g\left(\frac{1}{2}\right)$ is

(A) π (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. **[A]**

Sol. $g(a) = \lim_{h\to 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1}\,dt$

$$g\left(\frac{1}{2}\right) = \lim_{h\to 0^+} \int_h^{1-h} \frac{1}{\sqrt{t}\sqrt{1-t}}dt = \int_0^1 \frac{dt}{\sqrt{t}\sqrt{1-t}} = \int_0^1 \frac{dt}{\sqrt{t-t^2}}$$

$$= \int_0^1 \frac{dt}{\sqrt{\frac{1}{4} - \left(t - \frac{1}{2}\right)^2}} = \left[\sin^{-1}\left(\frac{t-\frac{1}{2}}{\frac{1}{2}}\right)\right]_0^1$$

$$= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

Q.54 The value of $g'\left(\frac{1}{2}\right)$ is

(A) $\frac{\pi}{2}$ (B) π (C) $-\frac{\pi}{2}$ (D) 0

Ans. **[D]**

Sol. Given that $g(a) = \int_0^1 \frac{dt}{t^a(1-t)^{1-a}}$

so $g\left(\frac{1}{2}-a\right) = \int_0^1 \frac{dt}{t^{\frac{1}{2}-a}(1-t)^{\frac{1}{2}+a}}$

$= \int_0^1 \frac{dt}{(1-t)^{\frac{1}{2}-a}.t^{\frac{1}{2}+a}}$ $\left[\text{using } \int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$

$= g\left(\frac{1}{2}+a\right)$

Hence $g\left(\frac{1}{2}-a\right) = g\left(\frac{1}{2}+a\right)$; i.e., g(x) is symmetric about x = 1/2.

differentiate it w.r.t. a

$-g'\left(\frac{1}{2}-a\right) = g'\left(\frac{1}{2}+a\right)$

using a = 0 in it

$-g'(1/2) = g'(1/2)$

so $g'(1/2) = 0$

Alternative Solution :

$g(a) = \int_0^1 \frac{dt}{t^a(1-t)^{1-a}} = \int_0^1 \left(\frac{1-t}{t}\right)^a \frac{1}{1-t}\,dt$

$g'(a) = \int_0^1 \frac{1}{1-t} \times \left(\frac{1-t}{t}\right)^a \ell n\left(\frac{1-t}{t}\right) dt$

$g'\left(\frac{1}{2}\right) = \int_0^1 \frac{1}{\sqrt{t(1-t)}} \ell n\left(\frac{1-t}{t}\right) dt$

Let $f(t) = \frac{1}{\sqrt{t(1-t)}}\ \ell n\left(\frac{1-t}{t}\right)$

$f(1-t) = \frac{1}{\sqrt{(1-t)t}}\ \ell n\left(\frac{t}{1-t}\right) = -f(t)$

$\therefore\ g'\left(\frac{1}{2}\right) = 0$

Paragraph For Questions 55 and 56

Box 1 contains three cards bearing numbers 1, 2, 3, box 2 contains five cards bearing numbers 1, 2, 3, 4, 5 and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the ith box, i = 1, 2, 3.

Q.55 The probability that $x_1 + x_2 + x_3$ is odd, is

(A) $\frac{29}{105}$ (B) $\frac{53}{105}$ (C) $\frac{57}{105}$ (D) $\frac{1}{2}$

Ans. **[B]**

Sol. as $x_1 + x_2 + x_3$ is odd

so all may be odd or one of them is odd and other two are even.

$\therefore$ Required probability

$$= \frac{{}^2C_1 \times {}^3C_1 \times {}^4C_1 + {}^1C_1 \times {}^2C_1 \times {}^4C_1 + {}^2C_1 \times {}^2C_1 \times {}^3C_1 + {}^1C_1 \times {}^3C_1 \times {}^3C_1}{{}^3C_1 \times {}^5C_1 \times {}^7C_1}$$

$$= \frac{24+8+12+9}{105} = \frac{53}{105}$$

Q.56 The probability that x_1, x_2, x_3 are in an arithmetic progression, is

(A) $\frac{9}{105}$ (B) $\frac{10}{105}$ (C) $\frac{11}{105}$ (D) $\frac{7}{105}$

Ans. **[C]**

Sol. x_1, x_2, x_3 are in A.P.

$$x_1 + x_3 = 2x_2$$

So $x_1 + x_3$ should be even number

either both are odd or both are even

$\therefore$ required probability $= \frac{{}^2C_1 \times {}^4C_1 + {}^1C_1 \times {}^3C_1}{{}^3C_1 \times {}^5C_1 \times {}^7C_1} = \frac{11}{105}$

SECTION – 3 : Matching List Type (Only One Option Correct)

This section contains four questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D), out of which one is correct.

Q.57 Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right)$; k = 1, 2, …, 9.

List I	List II
P. For each z_k there exists a z_j such that $z_k \cdot z_j = 1$	**1.** True
Q. There exists a $k \in \{1, 2, \ldots, 9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.	**2.** False
R. $\frac{\lvert 1-z_1\rvert\lvert 1-z_2\rvert\ldots.\lvert 1-z_9\rvert}{10}$ equals	**3.** 1
S. $1-\sum_{k=1}^{9}\cos\left(\frac{2k\pi}{10}\right)$ equals	**4.** 2

	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	3	4
(C)	1	2	3	4
(D)	2	1	4	3

Ans. **[C]**

Sol. Given $z_k = e^{i\frac{2k\pi}{10}}$

(P) $z_k.z_j = e^{i\left(\frac{2\pi}{10}\right)(k+j)}$

There exists $z_k.z_j = 1$

For example k = 1 & j = 9

(True)

(Q) $z = \frac{z_k}{z_1} = e^{i\left(\frac{2k\pi}{10}-\frac{2\pi}{10}\right)} = e^{i\frac{\pi}{5}(k-1)}$

for k = 2; $z = e^{i\frac{\pi}{5}}$

which in the given set (false)

(R) $\frac{|1-z_1||1-z_2|....|1-z_9|}{10}$

Note $|1-z_k| = \left|1-\cos\frac{2\pi k}{10}-i\sin\frac{2\pi k}{10}\right| = \left|2\sin\frac{\pi k}{10}\right|\left|\sin\frac{\pi k}{10}-i\cos\frac{\pi k}{10}\right| = 2\left|\sin\frac{\pi k}{10}\right|$

Now, Required product is

$$\frac{2^9 \sin\frac{\pi}{10}.\sin\frac{2\pi}{10}.\sin\frac{3\pi}{10}...\sin\frac{8\pi}{10}.\sin\frac{9\pi}{10}}{10} = \frac{2^9\left(\sin\frac{\pi}{10}\sin\frac{2\pi}{10}\sin\frac{3\pi}{10}\sin\frac{4\pi}{10}\right)^2 \sin\frac{5\pi}{10}}{10}$$

$$= \frac{2^9\left(\sin\frac{\pi}{10}\cos\frac{\pi}{10}.\sin\frac{2\pi}{10}\cos\frac{2\pi}{10}\right)^2.1}{10} = \frac{2^9\left(\frac{1}{2}\sin\frac{\pi}{5}.\frac{1}{2}\sin\frac{2\pi}{5}\right)^2}{10}$$

$$= \frac{2^5(\sin 36^\circ.\sin 72^\circ)^2}{10} = \frac{2^5}{2^2\times 10}(2\sin 36^\circ \sin 72^\circ)^2$$

$$= \frac{2^2}{5}(\cos 36^\circ - \cos 108^\circ)^2 = \frac{2^2}{5}\left[\left(\frac{\sqrt{5}-1}{4}\right)+\left(\frac{\sqrt{5}+1}{4}\right)\right]^2$$

$$= \frac{2^2}{5}.\frac{5}{4} = 1$$

(S) $1 + \alpha + \alpha^2 + \alpha^3 ..... + \alpha^9 = 0$

$$1 + \sum_{k=1}^{9}\alpha^k = 0$$

$$1 + \sum_{k=1}^{9}\left(\cos\frac{2k\pi}{10} + i\sin\frac{2k\pi}{10}\right) = 0$$

$$1 + \sum_{k=1}^{9}\cos\frac{2k\pi}{10} = 0$$

$$\text{So } 1 - \sum_{k=1}^{9}\cos\frac{2k\pi}{10} = 2$$

(P) $\rightarrow$ 1, (Q) $\rightarrow$ 2, (R) $\rightarrow$ 3, (S) $\rightarrow$ 4

Q.58

List I	**List II**
P. The number of polynomials f(x) with non-negative integer coefficients of degree ≤ 2, satisfying f(0) = 0 and $\int_0^1 f(x)dx = 1$, is	**1.** 8
Q. The number of points in the interval $\left[-\sqrt{13}, \sqrt{13}\right]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is	**2.** 2
R. $\int_{-2}^{2}\frac{3x^2}{(1+e^x)}dx$ equals	**3.** 4
S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}}\cos 2x \log\left(\frac{1+x}{1-x}\right)dx\right)}{\left(\int_{0}^{\frac{1}{2}}\cos 2x \log\left(\frac{1+x}{1-x}\right)dx\right)}$ equals	**4.** 0

	P	Q	R	S
(A)	3	2	4	1
(B)	2	3	4	1
(C)	3	2	1	4
(D)	2	3	1	4

Ans. **[D]**

Sol. (P) Let $f(x) = ax^2 + bx + c$

$f(0) = 0 \Rightarrow c = 0$

Now $\int_0^1 f(x)\,dx = 1 \Rightarrow \left[\frac{ax^3}{3} + \frac{bx^2}{2}\right]_0^1 = 1$

$\Rightarrow \frac{a}{3} + \frac{b}{2} = 1 \qquad \Rightarrow 2a + 3b = 6$

as a, b are non negative integers

so a = 0, b = 2 or a = 3, b = 0

so $f(x) = 2x$ or $f(x) = 3x^2$

(Q) $f(x) = \sin(x^2) + \cos(x^2) = \sqrt{2}\cos\left(x^2 - \frac{\pi}{4}\right)$

$\cos\theta$, attains maximum at $\theta = 0, 2\pi$

so $x^2 - \frac{\pi}{4} = 0, 2\pi$

$\Rightarrow x = \pm\sqrt{\frac{\pi}{4}}, \pm\sqrt{\frac{9\pi}{4}}$

so f(x) attains maximum at 4 points in $[-\sqrt{13}, \sqrt{13}]$

(R) $I = \int_{-2}^{2} \frac{3x^2}{1+e^x}\,dx = \int_0^2 \left(\frac{3x^2}{1+e^x} + \frac{3x^2}{1+e^{-x}}\right)dx = \int_0^2 3x^2\,dx = \left[x^3\right]_0^2 = 8$

(S) Let $f(x) = \cos 2x \log\left(\frac{1+x}{1-x}\right)$

$f(-x) = \cos 2x \log\left(\frac{1-x}{1+x}\right) = -f(x)$

hence f(x) is an odd function

so $\int_{-1/2}^{1/2} f(x)\,dx = 0$

(P) $\rightarrow$ 2; (Q) $\rightarrow$ 3; (R) $\rightarrow$ 1; (S) $\rightarrow$ 4

Q.59

List I	List II
P. Let $y(x) = \cos(3\cos^{-1}x)$, $x \in [-1,1]$, $x \neq \pm\frac{\sqrt{3}}{2}$. Then $\frac{1}{y(x)}\left\{(x^2-1)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ equals	**1.** 1
Q. Let $A_1, A_2, \ldots, A_n$ $(n > 2)$ be the vertices of a regular polygon of n sides with its centre at the origin. Let $\vec{a_k}$ be the position vector of the point A_k, $k = 1,2, \ldots, n$. If $\left\|\sum_{k=1}^{n-1}\left(\vec{a_k}\times\vec{a_{k+1}}\right)\right\| = \left\|\sum_{k=1}^{n-1}\left(\vec{a_k}\cdot\vec{a_{k+1}}\right)\right\|$, then the minimum value of n is	**2.** 2
R. If the normal from the point P(h, 1) on the ellipse $\frac{x^2}{6}+\frac{y^2}{3} = 1$ is perpendicular to the line $x + y = 8$, then the value of h is	**3.** 8
S. Number of positive solutions satisfying the equation $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$ is	**4.** 9

	P	Q	R	S
(A)	4	3	2	1
(B)	2	4	3	1
(C)	4	3	1	2
(D)	2	4	1	3

Ans. **[A]**

Sol. **(P)** $y(x) = \cos(3\cos^{-1}x)$

$$y'(x) = \sin(3\cos^{-1}x)\frac{-3}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2}\ \frac{dy}{dx} = -3\sin(3\cos^{-1}x)$$

$$\Rightarrow \frac{-x}{\sqrt{1-x^2}}\frac{dy}{dx}+\sqrt{1-x^2}\ \frac{d^2y}{dx^2} = -3\cos(3\cos^{-1}x)\frac{(-3)}{\sqrt{1-x^2}}$$

$$\Rightarrow -x\frac{dy}{dx}+(1-x^2)\frac{d^2y}{dx^2} = -9y(x)$$

$$\Rightarrow \frac{1}{y(x)}\left[(x^2-1)\frac{d^2y}{dx^2}+x\frac{dy}{dx}\right] = 9$$

(Q) $\vec{a}_k \times \vec{a}_{k+1} = |r_1||r|\sin\frac{2\pi}{n}\hat{n}$

$$\left|\sum_{k=1}^{n-1}\vec{a}_k \times \vec{a}_{k+1}\right| = \left|(n-1)\sin\frac{2\pi}{n}\hat{n}\right|$$

$$\left|\sum_{k=1}^{n-1}\vec{a}_k \cdot \vec{a}_{k+1}\right| = \left|(n-1)\cos\frac{2\pi}{n}\right|$$

Both are equal

$$\left|\sin\frac{2\pi}{n}\right| = \left|\cos\frac{2\pi}{n}\right| \Rightarrow n_{min} = 8 \text{ (for polygon } n \geq 3)$$

(R) Equation of normal is

$\sqrt{6}\,x\sec\theta - \sqrt{3}\,y\,\text{cosec}\theta = 3$

Its slope is $\frac{\sqrt{6}\sec\theta}{\sqrt{3}\,\text{cosec}\theta} = 1$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$

So normal is

$\sqrt{6}\;x\;\frac{\sqrt{3}}{\sqrt{2}} - \sqrt{3}\times\sqrt{3}\;y = 3$

$3x - 3y = 3 \Rightarrow x - y = 1$

as it passes through (h, 1)

So $h - 1 = 1$

$h = 2$

(S) $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{2x+1}+\frac{1}{4x+1}}{1-\frac{1}{(2x+1)(4x+1)}}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{6x+2}{(2x+1)(4x+1)-1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$$

$$\Rightarrow \frac{3x+1}{4x^2+3x} = \frac{2}{x^2} \qquad \Rightarrow 3x^3 + x^2 = 8x^2 + 6x$$

$\Rightarrow x(3x^2 - 7x - 6) = 0$

$\Rightarrow x(x-3)(3x+2) = 0$

$\Rightarrow x = 0, \frac{-2}{3}, 3$

So only Positive solution is $x = 3$

Q.60 Let $f_1 : R \to R$, $f_2 : [0, \infty) \to R$, $f_3 : R \to R$ and $f_4 : R \to [0, \infty)$ be defined by $f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$

$f_2(x) = x^2$; $f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$ and $f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0 \end{cases}$

List I	**List II**
P. f_4 is	**1.** onto but not one-one
Q. f_3 is	**2.** neither continuous nor one-one
R. $f_2 o f_1$ is	**3.** differentiable but not one-one
S. f_2 is	**4.**continuous and one-one

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

Ans. **[D]**

Sol. $f_1(x) = \begin{cases} -x, & x < 0 \\ e^x, & x \geq 0 \end{cases}$

$f_2(x) = x^2, \ x \geq 0$

$f_3(x) = \begin{cases} \sin x, & x < 0 \\ x, & x \geq 0 \end{cases}$

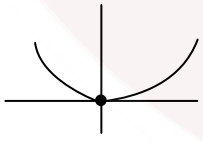
$f_4(x) = \begin{cases} f_2(f_1(x)), & x < 0 \\ f_2(f_1(x)) - 1, & x \geq 0 \end{cases}$

Now $f_2(f_1(x)) = \begin{cases} x^2, & x < 0 \\ e^{2x}, & x \geq 0 \end{cases}$

(P) $f_4 = \begin{cases} x^2, & x < 0 \\ e^{2x} - 1, & x \geq 0 \end{cases}$

as f(x) is continuous

$f_4'(x) = \begin{cases} 2x, & x < 0 \\ 2e^{2x}, & x > 0 \end{cases}$

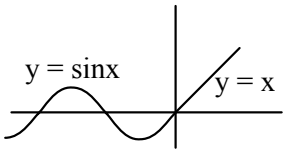


f′(0) is not defined

its range is $[0, \infty)$

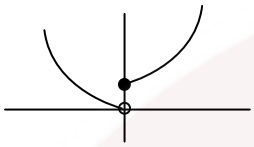
So it is onto but not one-one

(Q) $f_3(x)$



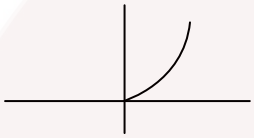
differentiable at x = 0 and not one-one

(R) $f_2(f_1(x))$



it is neither continuous nor one-one

(S) $f_2(x)$



It is continuous and one-one.

$P \to 1,\ Q \to 3,\ R \to 2,\ S \to 4$